ECE 501 - DIGITAL IMAGE PROCESSING

Explainable Multi-Label Content-Based Image Retrieval for Aerial Images using Classical Computer Vision

Weekly Progress Report

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1 Introduction

During Week 6, the team implemented the first half of Phase 5 of the Content-Based Image Retrieval (CBIR) system: validation-based tuning of the feature-fusion weights (Phase 5A) and the explainability module (Phase 5B). The work follows the plan made in Week 5. The weights are learned on a validation split that is kept separate from the test queries, and every retrieved image is accompanied by a per-cue score breakdown and region-based heatmaps.

The uncertainty analysis and the final performance evaluation (Phases 5C and 5D) are scheduled for Week 7, so that the remaining Phase 5 work is divided equally between the two weeks.

2 Review of Existing System

The Phase 4 system combines four cues (RGB colour histogram, Gabor texture, HOG shape and grid colour moments) with equal weights of 0.25. On the 600 held-out queries against the 2,400-image gallery it reaches 50.10% same-class Precision@5, 78.50% Hit@5, 25.55% mAP and 84.67% multi-label Precision@5 (Jaccard ≥ 0.5).

Adding the spatial cue to the colour, texture and shape fusion with equal weights slightly lowered Precision@5 (from 50.97% to 50.10%), which indicated that equal weights are not the best combination. Before starting Phase 5, the dataset was set up again and the Phase 4 results were reproduced exactly from the cached features (50.10% P@5, 25.55% mAP), confirming that the pipeline used for Phase 5 is identical to the earlier phases.

3 Phase 5A: Weight Tuning and Score Fusion

3.1 Visual cues and similarity scores

Every descriptor from Phases 1 to 4 is treated as an individual cue, giving 11 cues: RGB and HSV histograms, LBP, GLCM and Gabor texture, HOG, edge histogram and Hu moments, grid colour moments, grid intensity and the colour layout descriptor (CLD). The similarity of each cue is computed independently with the conventions of the earlier phases (histogram intersection for histogram descriptors, gallery z-score cosine similarity mapped to [0, 1] for vector descriptors) and stored as a matrix, so that different weight vectors can be tested quickly.

3.2 Fusion model and validation protocol

$$S(q, g) = \Sigma_c \, w_c \cdot S_c(q, g), \; w_c \geq 0, \; \Sigma_c \, w_c = 1$$

The 600 validation images (20 per class, fixed random seed) are used as queries against the other 1,800 gallery images, so the test queries are never used for tuning. The objective is the average, over five relevance definitions (same scene class and Jaccard thresholds 0.5, 0.6, 0.7 and 0.8), of the mean of mAP and Precision@10. The weights are optimised with NumPy only: random starting points on the weight simplex followed by pairwise weight transfer (coordinate ascent) with a shrinking step. Three random seeds were run for the 11-cue case to check stability.

3.3 Results

Tuning only the four Phase 4 cues gave almost no change (50.23% P@5 compared with 50.10%). Tuning all 11 descriptors raised same-class Precision@5 on the held-out queries from 50.10% to 56.50% and mAP from 25.55% to 29.24% (Table 1). The learned weights (Figure 1) give 0.349 to LBP and 0.205 to HOG, while Hu moments receive zero weight. By modality the weights are texture 0.43, shape 0.24, spatial 0.17 and colour 0.16. The weights were stable across the three seeds (standard deviation at most 0.022). The gain therefore comes from using the stronger texture descriptors (LBP, GLCM) that the four-cue system left unused, not from re-weighting the four original cues.

Table 1: Effect of weight tuning on the 600 held-out test queries (%).

System	Same-class P@5	Same-class mAP	Jaccard ≥ 0.5 P@5	Jaccard ≥ 0.5 mAP
4-cue equal (Phase 4)	50.10	25.55	84.67	64.55
4-cue tuned	50.23	25.31	84.50	65.69
11-cue equal	50.57	25.70	84.53	66.05
11-cue tuned (final)	**56.50**	**29.24**	**86.23**	**66.73**

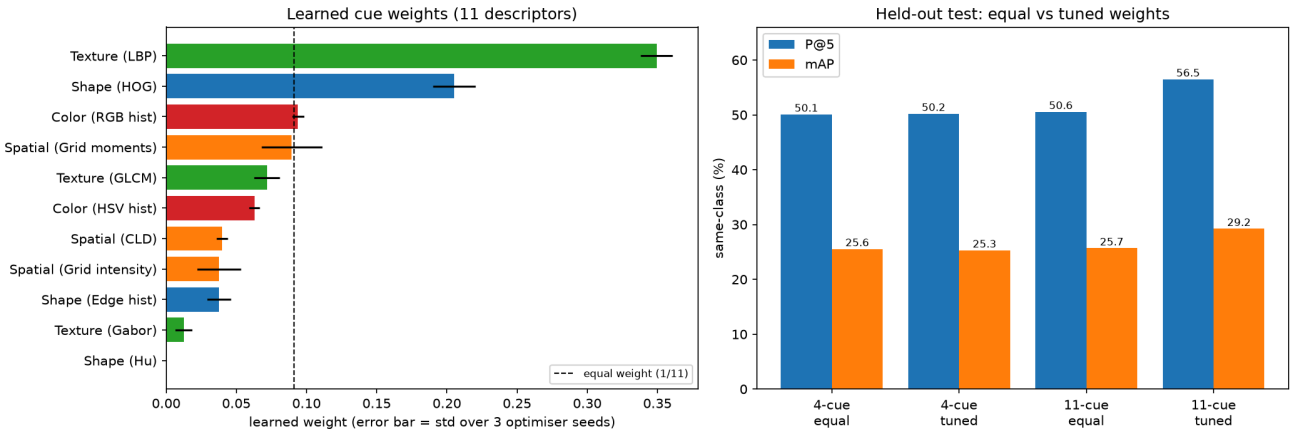


Figure 1: Learned cue weights (left) and equal versus tuned weights on the test queries (right).

4 Phase 5B: Explainability

4.1 Per-cue score breakdown

Because the fusion is linear, the final similarity score splits exactly into the contributions $w_c S_c$ of the individual cues. For each retrieved image the system reports these contributions and also the percentile of each cue score within the gallery, since the same raw value means different things for a histogram intersection and a cosine similarity. For the query airport_85 and its rank-1 match airport_22, LBP contributes 41%, HOG 18% and grid colour moments 10% of the score.

4.2 Region-based heatmaps

The heatmaps use occlusion on the real pipeline. Both images are divided into an 8 × 8 grid. For each cell, the same cell is blanked in both images, all 11 descriptors are extracted again with the original extractors and the pair is re-scored. The drop in similarity shows how much that region supports the match. The first version gave maps that were almost entirely negative, because blanking the same cell in both images adds an identical flat patch, which makes any pair look more alike. Each map is therefore centred on its mean cell effect, and the final maps show whether a region supports the match more or less than an average region. Maps are also produced per modality (colour, texture, shape, spatial) by summing the weighted maps of its descriptors (Figure 2).

4.3 Check of the heatmaps

A deletion test was run on the rank-1 pair of one test query from each of the 30 classes. Blanking the 4 of 16 cells that the map marks as most supportive changed the fused similarity by −0.069 on average, blanking 4 random cells by −0.083 and blanking the 4 least supportive cells by −0.098. The ordering held in all 30 pairs, so the maps follow the score. The absolute values are negative because of the flat-patch effect described above. The maps explain which regions change the score, not what the scene contains, and the colour and texture maps are smoother because those cues are global statistics.

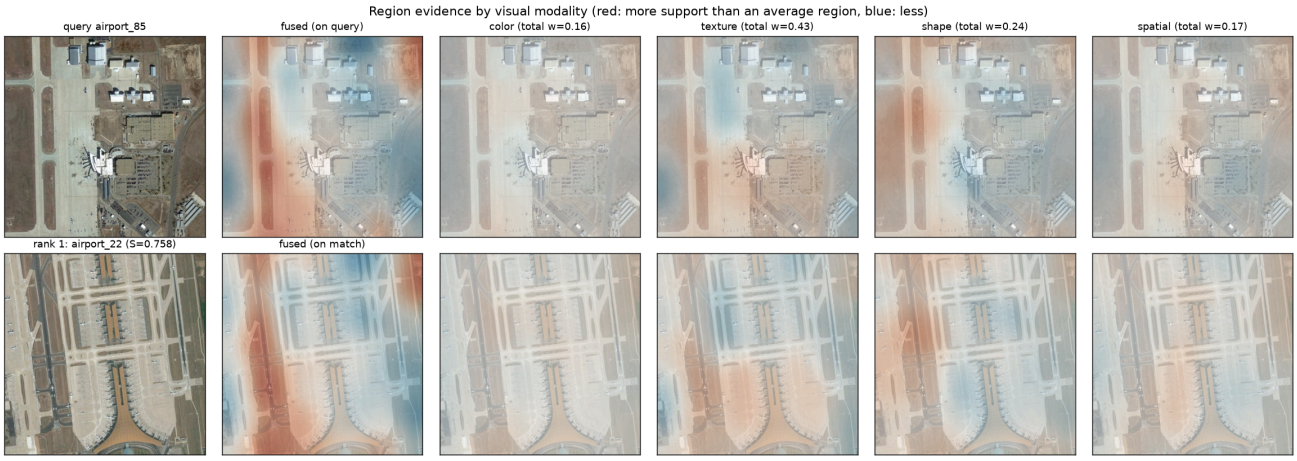


Figure 2: Region evidence for query airport_85 and its rank-1 match, fused and per modality (red: more support than an average region, blue: less).

5 Work Completed

During Week 6, the following work was carried out:

- Set up the dataset and reproduced the Phase 4 results exactly from the cached features.
- Computed and cached the similarity matrices of all 11 descriptors for the validation and test protocols.
- Implemented the validation protocol and the NumPy weight optimiser, and tuned the weights for the 4-cue and 11-cue settings.
- Evaluated tuned and equal weights on the held-out queries and analysed the learned weights.
- Implemented the per-cue score breakdown with gallery percentiles.
- Implemented the 8 × 8 occlusion heatmaps, corrected their bias by centring, and added per-modality maps.
- Ran the deletion test on 30 query-match pairs and generated explanation figures for all 30 classes.

6 Plan for Next Week

The next stage will complete Phase 5 and combine the components into one system.

- Implement uncertainty indicators based on score margins and disagreement between cues.
- Build and test reliability flags without using the labels of the test queries.
- Evaluate the complete system with Precision@K and mAP at Jaccard thresholds from 0.5 to 0.8 and compare it with Phase 4.
- Combine ranking, explanations and flags in a single final system and generate outputs for all 30 classes.

7 Conclusion

Week 6 completed Phases 5A and 5B. Learning the fusion weights on a separate validation split improved same-class Precision@5 from 50.10% to 56.50% on the held-out queries, with most of the weight placed on LBP and HOG. The retrieval results now come with a per-cue score breakdown and region heatmaps, and a deletion test supports the heatmaps. The remaining work is the uncertainty analysis, the final evaluation and the integration of all components.